Gains Hide Failures: A Ledger Evaluation of a Commitment Guard for AAC Sentence Expansion

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Abstract

People who use AAC (augmentative and alternative communication) pick a few words, and a language model writes the sentence the app speaks, sometimes dropping a "not". We evaluate a commitment *guard*, which makes the model rewrite when its sentence contradicts the user's input, in a *ledger* that sets each gain beside what it leaves broken, what it breaks and what it costs. All results are descriptive stored-text contrasts, with no effect on people. On 78 frozen-guard speech-act families, the guard raises the joint pass rate by +9.83 points [8.39, 11.32], repairs 52.51% of strict-check failures, and changed 26 already-correct texts, 0 of which became failures. In a human audit of 120 texts with the rule frozen in advance, the frozen rule returns no detectable usability cost for repairs against never-failed controls (71.7% against 85.0%, −13.3 points [−27.6, +1.5]), the deployment clause is unresolved, and stricter endpoints show a cost. Models commit to the affirmative reading in at least 98.4% of committed answers even when told not to assume, so their rare errors are silent reversals; a supplied cue, as a chip or a plain word, removes nearly all of them in same-item regeneration. Writers accept 1 of 200 outputs as written, so the ledger reports whole-message fidelity beside the commitments, with post hoc checks of earlier predictions and a pre-specified human-effect design.

1 Introduction

People who use AAC often compose by selecting a few words or symbols, because every selection costs time and effort (Valencia et al. 2023; Cai et al. 2024). LLM keyword expansion turns `want go party` into a full sentence (Shen et al. 2022; Cai et al. 2024), which a system may speak on the user's behalf, sometimes after review (Hancock, Naaman, and Levy 2020; Valencia et al. 2020). But the model decides more than words: whether something happened or did not, whether the sentence asks or tells, whether it says "please". Given `not like medication`, a sentence that drops the negation, turns it into a question or adds politeness is fluent and wrong. Three kinds of fix are being explored: check the output against the user's words and regenerate on a conflict (Madaan et al. 2023; Gou et al. 2024), offer suggestions that span the readings the words leave open, or give the user a control over stance or context (Valencia et al. 2024; Mao et al. 2025; Xiang et al. 2025), which our one-tap cue instantiates. Such work commonly reports accuracy or similarity gains (Cai et al. 2022; Shen et al. 2022; Cai et al. 2024), which are real, but a gain-only report hides what a fix leaves unfixed, what it changes that was already right, and whether the repaired text is usable.

User input	Model wrote	Guard output	What the scorer and the person see
`[?] [I] bring friend`	"Is it okay if I bring a friend?"	"Is [I] bring a friend?"	Scored a pass (it keeps the question label). Not a sentence: the model copied the chip token.
`knock come`	"Please knock before you come in."	"Don't knock before you come in."	A menu meant to widen choices turned a polite request into a prohibition.
`[ASK] [YOU] [PLAIN] wear headphones`	"You wear headphones."	"Ask you to wear headphones."	Scored a pass. Not what was meant.

Figure 1: Illustrative failures found by the audit, real stored outputs. Bracketed tokens in an input are cue chips, single taps that fix one commitment: `[NOT]` negation, `[?]` a question, `[ASK]` and `[YOU]` an ask addressed to the listener, `[I]` the speaker as agent, `[PLAIN]` no added politeness. They are not typical: each is the first case in a fixed hash order within its category, except row 3, the second repair in that order; rates are in Section 6.2. Row 1: the guard changed 26 of 1,394 already-correct texts, none into a strict-check failure. Row 2: menu rewrite; sets offering both readings fell from 26/40 to 12/40.

Thesis and ledger. Where only gains are reported, failures stay hidden, so we report a ledger. Beside each benefit it records *offline failure indicators* (events in stored text that a gain-only report omits: failures that remain, correct outputs the fix changed, readings a menu no longer offers, silent re-

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versals under a default that is right on average), the cost, and where the intervention applies. None of these indicators is a measured harm to a person. **Does it work?** Yes, for what it was built to protect: the guard repairs supplied-value failures on fresh items, against their own failed originals its repairs keep the meaning and remove invented stance, the frozen usability rule returns no detectable cost (the deployment clause is unresolved and stricter endpoints show a cost), and a supplied cue removes nearly all silent reversals; the open column is the whole message, which the ledger now measures. Our running case is keyword-to-sentence expansion with four LLMs on minimal-pair item families, defined in Section 3.

Contributions. (1) A ledger and its design rules: attribution decides what an intervention may be credited with; a free baseline and pass lines fixed in advance decide whether it merits a human study (§5). (2) One lead case, a commitment guard, evaluated end to end: on a confirmation set with a frozen rule it raises the joint pass rate by +9.83 points, repairs 52.51% of strict-check failures (46.3% of joint-pass failures), rewrote 26 of 1,394 already-correct texts with none becoming a strict-check failure, and its feedback beats plain retrying by +6.94 points [5.34, 8.60] at matched decoding (+6.20 against resampling at temperature 0.7); the ledger adds residual failures, the share a free baseline earns, and a 160-text human audit of its repairs (§6). (3) Two more ledger rows (a menu, and a cue whose gain replicates but whose confirmation misses its 80%-of-typed line), realistic-input checks, and an analysis of the affirmative default as a good guess with an asymmetric cost. (4) Three columns to report beside commitment retention: writer-intent match, clinically weighted failure, and the measured price of a missing cue (§6.5, §6.6). (5) A pre-specified human-effect design with falsifiable predictions and the design implications of two AAC-user consultants (§7). We ran no study with AAC users; every result is a change in stored text.

2 Related Work

LLMs in AAC. LLM expansion and suggestion systems for AAC include KWickChat (Shen et al. 2022) and SpeakFaster (Cai et al. 2024), and AAC users have described how such systems help and hinder them (Valencia et al. 2023). Several add user-side control over stance or context (Valencia et al. 2024; Mao et al. 2025; Xiang et al. 2025), one keeps a short list of bracketed function words, including `not`, in the abbreviation (Lacosse, Tano, and McNamee 2026); others measure whether affect or intent survives generation (Choudhury, Kumar, and Martin 2025). Frisch et al. argue that AI-powered AAC needs technical, user-mediated and functional evaluation together (Frisch et al. 2026); our ledger is a technical layer shaped so that a functional study can test it.

Editing, preservation and evaluation. Generate–critique–revise loops (Madaan et al. 2023; Gou et al. 2024) and conditioning a rewrite on the source dialog act (Qu, Ungar, and Sedoc 2023) are established, as is reporting a targeted change together with collateral change (Ziegenbein, Stahl, and Wachsmuth 2026; Elgaar and Amiri 2025). Preservation metrics are valid only when test data do not confound the intended change with content errors (Pauli, Augenstein, and Assent 2025); models have default politeness strategies (Zhao and Hawkins 2025), and write sentences wrongly positive about negative commonsense facts (Chen et al. 2023), so an affirmative default is not our finding; we price it. We add accounting conditioned on what the input *supplied*, against a free baseline, plus a human usability check on repaired text. Suggested text shifts what people write (Arnold, Chauncey, and Gajos 2020; Jakesch et al. 2023; Hohenstein et al. 2023; Agarwal, Naaman, and Vashistha 2025); we screen interventions meant to prevent that. LLM judges (Zheng et al. 2023) favor their own generations (Panickssery, Bowman, and Feng 2024) and can agree with each other more than with people on pragmatic labels (Wang, Sun, and Guo 2026), which bounds what our machine labels can show.

3 Setting

Items and populations. An item is a minimal-pair family ("I don't like the new medication" / "I like the new medication"); each source sentence is shown to the model under five input conditions (Table 1). The *development set* has 58 constructed families on six dimensions (after quarantining two families whose scoring rule has a known blind spot); the guard was built and tested there, so its numbers are development results, and the menu and cue rows use it. The *confirmation set* has 240 fresh constructed families (polarity, speech act, politeness), 238 analyzed after a pre-generation exclusion, with three contrasts pre-specified in version control; a hash of the protocol was deposited publicly after the confirmation freeze, so it does not certify the freeze. 48 cached requests predate the final item freeze, which qualifies it. Its 78 analysed speech-act families, a pre-specified subset of the 238 chosen because speech act was the guard's weakest development dimension, are where the guard was rerun unchanged. The *human-written pairs* are 547 kept minimal pairs written by 20 non-author annotators. The *keyword study* has 20 annotators type keywords for 600 AACText messages (crowd-written fiction, not AAC users' messages).

Scoring and attribution. Four primary models write each sentence: Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct and Gemma-3-4B-it (4-bit, llama.cpp) and one commercial API model; Section 6.4 adds Qwen2.5-14B and Mistral-Nemo-12B (six models from five families) and a lite Gemini model on the legend arms only (seven in all); the models are fixed, not sampled. An LLM extractor (two prompt variants) labels each output's dimensions without seeing the source, model or condition; a second-family arbiter resolves disagreements on critical labels; deterministic code turns labels into verdicts, identically for treated and untreated outputs. A target is one (item, model, input condition) outcome. Because the input condition fixes which values the model was given, a miss is attributed to one of three causes: *loss* (the input never carried the value), *violation* (it carried the value and the output changed it) or *assumption* (the input was ambiguous and the output committed anyway). Take the source "I don't like the new medication". The keywords `like medication` never carried the negation, so an affirmative output is a *loss* at input, and the remedy is a cue;

Glossary			
Family	Two source sentences differing on one critical dimension, with the same bare keywords	*Cue, chip*	A commitment the user adds, as one tap such as `[NOT]`, `[?]`, `[PLAIN]`, or as a plain word
Critical dimension	Polarity, agent, certainty, speech act (ask, tell, request), politeness, time/quantity	*Input conditions*	Keywords only; keywords plus a length-matched neutral word; plus the critical cue; plus all cues; the typed sentence ("typed input", the full channel)
Pass (joint pass)	Keeps the critical value and every other scored supplied predicate; the headline metric	*First pass*	The model's single greedy sentence, before any intervention
Strict-check pass	Keeps the critical value, contradicts no supplied dimension, adds no politeness; a failure is a *strict-check failure*, and *repair rate* is the share of these that the intervention fixes	*Strip-only*	Free baseline: delete an unrequested "please", "pls" or "plz" (no model call)
Violation	The input carried a value and the output changed it	*Offline failure indicator*	Event in stored text that a gain-only report omits

Table 1: Glossary. All scores are partial predicates computed from machine-extracted labels, not judgments of usability.

`[NOT] like medication` carried it, so an affirmative output is a *violation*, and only then can a checker enforce the value; when the words leave both readings open and a menu of suggestions offers only one, the miss is an *assumption*, and the remedy is a menu or a question. The guard is therefore credited with violations alone, and a gain is never credited to an intervention for a failure of a kind it cannot touch. In the confirmation set, 259 disagreement rows (177 units) were resolved by a single arbiter after outcomes were known, blind to reference, model and condition. Against two human annotators on 163 sampled untreated outputs, extractor labels agree at held-out $\kappa = 0.91$–0.97 on all 7 label fields (pre-specified gate 0.6). In a check of arbitrated labels by 20 annotators (units where the two judges disagreed), people sided with the arbiter in 77% of codings ($\kappa = .74$) against 60% and 40% for the two judges, and 98% coded "Can you. . . ?" as a request. Repaired text is checked by the usability audit (Section 6.2). An anonymous concurrent submission (Anonymous 2027) describes the item construction and scorer and shares data with this paper; the guard and ledger results are reported here in full.

4 Interventions

The commitment guard (lead; system side). After the first pass, a deterministic lexical monitor (regexes and lexicons for negation, person, modality, question form, numbers, tense and "please") compares the output with the values in the visible input: the cues, or for typed input the monitor's own labels of the typed sentence. It also flags a politeness marker that was not supplied. On a flag, the same conversation receives one feedback turn ("Your sentence must keep: negated; no added politeness. Write the sentence again with the same meaning otherwise.") and is regenerated, up to two retries. The guard returns the first unflagged attempt, else the attempt with fewest flags, ties going to the original. The free baselines are strip-only (one failure type) and plain retrying. The no-feedback baseline (G1r) keeps the monitor, trigger and retry cap but regenerates from the original prompt with no feedback turn, at temperature 0.7 because a temperature-0 retry repeats the first pass; the guard at the same temperature (G1t) isolates the feedback message.

The menu (secondary; keywords only). The complete menu keeps the top of three baseline suggestions and rewrites the third (then second) into a missing polarity (then certainty) reading, with one model call and one monitored retry.

The cue (secondary; user side). The user adds one commitment cue, such as `[NOT]`, `[MAYBE]`, `[?]` or `[PLAIN]`. The chip probes whether supplied information is used; the lever is the missing cue, which a plain word can also supply (Sec. 6.4).

5 Evaluation Design

Evaluation card: intervention, comparison, pass line. Pass lines were fixed in advance (development: in code before the run; confirmation: in a hashed file before any retry); verdicts are in Table 2 and Section 6.2. *Guard* (targets a violation of a supplied value; compared with the stored first pass and with strip-only at zero calls): repair $\geq$30% of failures *and* change $\leq$10% of correct outputs; confirmation pass-rate gain with lower bound > 0; deployment clause, usable $\geq$ control -10 points. *Menu* (an assumption under ambiguous input; compared with the stored baseline menu): gain $\geq$30 points. *Cue* (a loss of a value at input; compared with a length-matched neutral word): gain $\geq$10 points *and* $\geq$80% of typed-input rate.

Estimand, arms and what the ledger predicts. The human target (Section 7) is the excursion effect, on a proximal "said what I meant" outcome and on weekly longitudinal rates, of delivering a regenerated sentence instead of the original at a flagged decision point; the arms are silent regeneration, flag-and-confirm and the original. The offline columns are the predictors tested against it: *changed-correct* should lower the proximal match rate and *coverage loss* (readings a menu no longer offers) the longitudinal rates; *reliance* (acceptance of flagged text rising over weeks) offline text cannot show.

Paired comparisons of stored configurations. For system-side interventions, the untreated outcome is the *stored*

first-pass output for the same request, and the treated outcome is what the intervention returns from it. This removes between-request variation but does not identify an effect on people: every "gain", "increment" or "effect" below is a descriptive contrast on fixed items, with no estimand for people (that estimand is the one above). Intervals are 95% percentile intervals from whole-family or cluster resamples (fixed seeds), pointwise and uncorrected for selection or label error.

Offline failure indicators and cost. *Changed-correct* counts already-correct outputs whose text the intervention changed; *correct→failure* is the share that became scored failures; *residual* failures remain after the intervention. We also show what a label cannot see (scored passes that are fragments; a human usability rating of repaired text). Cost is extra model calls and latency.

6 Results

6.1 The guard: about half of failures repaired, and feedback beats plain retrying

On the development cue conditions, strict-check pass rose from 94.25% to 98.78% of 1,392 targets (Table 2). Of the 80 initial failures the guard repaired 63 (78.8% [68.4, 88.9]), far above its 30% pass line. Deleting an unrequested "please" at zero calls already reaches 1,346 strict-check passes, so regeneration's increment is 2.08 points [1.15, 3.16]: 31 targets pass only with the guard and 2 only with deletion. 17 of 80 failures remained, 12 of them involving speech act, and 14/17 were still flagged after both retries: detected, not repaired. Regenerating without feedback repairs little: 15/80 development failures (18.8%) against 63/80 with the guard, and 87/478 confirmation failures (18.2%) against 251/478. The guard's increment over resampling (unmatched decoding) is +3.23 [1.87, 4.81] points in development and +6.20 [4.75, 7.64] in confirmation; the matched-decoding contrast (G1t − G1r) is the clean feedback contrast: +2.95 and +6.94 [5.34, 8.60].

Confirmation. Speech act was the guard's weakest dimension, so we reran the guard unchanged on the confirmation set's speech-act families (78 families, four models, three cue conditions, 1,872 targets). Criteria were frozen before any retry: at least 30% of strict failures repaired, at most 10% of strict-correct texts changed, and a joint-pass gain whose lower 95% bound exceeds zero. The pass rate rose from 78.79% to 88.62%, 9.83 points [8.39, 11.32]; the guard repaired 251/478 strict-check failures (52.51% [48.36, 56.70]), changed 26/1,394 already-correct texts and turned 0 of the strict-check passes into failures. Only the 26 changed texts could have failed, so the informative figure is 0 of 26 (95% upper bound 13%); the strict predicate shares lexical logic with the monitor and the guard returns its first unflagged attempt, so the zero is partly by construction. The shown row 1 is not such a failure by the scorer's labels; a post hoc lexical rule finds a bracketed chip token in 1/26 of the changed texts, 3/1,645 of all guard strict-check passes and 2/1,394 of first-pass passes, and a person is needed for malformed text of other kinds (Section 6.2). Every scored criterion passed within the call caps. The gain is uneven across the four fixed models: +0.85 (API model), +3.21 (Qwen), +16.24 (Llama) and +19.02 points (Gemma), 468 targets each, so the pooled figure comes mostly from the two weaker models. The labels behind every treated-output number were validated on untreated outputs only; no human relabelled guard-treated text beyond the 60 repaired texts of the audit below. The repair rate fell from 78.8% (strict-check; 78.4% joint) in development to 52.51% (46.3% joint) on fresh items, so the development figure overstates it; joint-pass and strict-check rates are not interchangeable. Deletion alone gained 3.47 points, so the guard's increment over it is 6.36 points [5.29, 7.48], and 12.13% of targets remain strict-check failures. Of the results here, the guard's three frozen confirmation criteria are confirmatory; the resample baseline, the audit's secondary endpoints, H1 and H2, the decomposition and the frontier rows are post hoc or exploratory.

6.2 What the pass does not show: repair usability

A pass checks a few labels, not a sentence. For `[ASK] [YOU] [PLAIN] wear headphones`, the guard turns "You wear headphones." into "Ask you to wear headphones.", which passes every scored predicate (Figure 1). The pattern is common: 4 of the 63 scored development repairs begin "Ask you" (post hoc lexical count). We therefore had people judge repaired text. **The audit.** A 160-text audit of new texts (120 primary: 60 repaired, 60 never-failed controls; 40 secondary: 20 original-failed and 20 strip-only texts, to separate repair from item difficulty) gave each text 3 independent non-author raters, under a rule frozen before any return: a pooled interval entirely below −10 points kills the guard for deployment (kept as a detector); one including −10 but excluding 0 is a cost of uncertain size; one including 0 is no detectable cost. Repaired texts were usable in 43/60 texts (71.7%) against 51/60 (85.0%) for controls, a gap of −13.3 points [−27.6, +1.5]. The frozen rule returns *no detectable usability cost* (the interval includes zero), the deployment clause is unresolved (the point estimate lies below the −10 line), and stricter endpoints show a cost. The audit can detect a gap of about 18 points near the observed control rate, so no detectable cost is not evidence of no cost, and only this usable endpoint was the frozen decision rule. The texts are a deterministic, hash-ordered, model-balanced list from the discovery and transfer pools, not a probability sample, so intervals describe these texts; controls are not item-matched to the repairs, so the gap mixes repair and item difficulty; by pool the usable gap is −16.7 [−35.6, +3.3] (discovery) and −10.0 [−30.8, +11.8] (transfer). Two stricter secondary endpoints, defined before the returns and uncorrected for multiplicity, do show a cost against never-failed controls: meaning preserved 79.7% against 93.2% (−13.6 points [−26.2, −1.0]) and usable *and* preserving 61.7% against 81.7% (−20.0 points [−34.8, −3.9]). The cost is uneven by model (descriptive, an unbalanced list: the smallest model supplied 19 repairs and the API model 12): usable in 6 of 19 and 12 of 12 texts. The intervals above treat items as independent although each item has 3 raters among 6 annotators and items cluster by family; resampling the 95 families gives −13.3 points [−28.6, +1.0]. Standardizing to a common model mix

	Development	**Confirmation (frozen; 48 early cached requests)**
	58 families, 6 dimensions, 1,392 cue targets	78 speech-act families (of 238), 1,872 targets
Benefit (% of targets)		
Strict-check pass: first pass / strip-only / guard	94.25 / 96.70 / 98.78	74.47 / 81.73 / 87.87
Pass (joint): same order	94.68 / 96.77 / 98.85	78.79 / 82.26 / 88.62
Guard − first pass, points	+4.17 pass; +4.53 strict-check	**+9.83** [8.39, 11.32] pass; +13.41 [11.38, 15.44] strict-check
Failures repaired, strict-check / joint pass	63/80 (78.8%) / 58/74 (78.4%)	251/478 (52.51%) / 184/397 (46.3%)
Free baseline: strip-only, zero calls (+2.44 strict-check development, +3.47 [2.78, 4.22] pass confirmation)		
Guard − strip-only, points	**+2.08** [1.15, 3.16] strict-check	+6.36 [5.29, 7.48] pass
Free baseline 2: resample without feedback (G1r), T=0.7, same retry cap; G1t is the guard rerun at T=0.7		
Pass: resample G1r / guard G1t (T=0.7)	95.6 / 98.6	82.4 / 89.4
Repair rate: resample G1r	15/80 (18.8%)	87/478 (18.2%)
Guard (T=0) − G1r, points (pass)	**+3.23** [1.87, 4.81]	**+6.20** [4.75, 7.64]
G1t − G1r, matched decoding (pass)	+2.95 [1.65, 4.38]	+6.94 [5.34, 8.60]
What remains (offline failure indicators)		
Residual strict-check failures	17/1,392 (1.2%)	227/1,872 (12.13%)
Already-correct text changed	14/1,312 cue; 49/2,154 all inputs	26/1,394 (1.87%)
Strict-check pass → failure	0/1,312	0/1,394; 0/26 changed (upper bound 13%)
Cost and verdict		
Audit vs controls (points; usable / preserving / both)	—	−13.3 [−27.6, +1.5] / −13.6 [−26.2, −1.0] / −20.0 [−34.8, −3.9]
Extra calls per input	0.104 (1,856 inputs)	0.41 (1,848 inputs)
Pass line	passed	passed on the three scored criteria (usability: §6.2)

Visible input	First pass	Guard output
Development set, repaired (Gemma, one cue; 1st by hash order; 63/80 scored failures repaired)		
`[PLAIN] let finish sentence`	"Please let me finish my sentence."	"Let me finish the sentence."
Development set, failure that remains (Llama, all cues; 1st by hash order; 17/80 failures remain, 14/17 still flagged; the source asks whether the listener is charging it)		
`[?] [YOU] charge wheelchair`	"Can you charge my wheelchair?"	"Can you charge my wheelchair?"

Table 2: The guard as one display. All contrasts are descriptive stored-text contrasts, with no effect on people. Left: benefit, the free baseline and what remains, on two *separate* populations that are never pooled; scored partial predicates, not usability. *Targets* are scored (item, model, input condition) outcomes; *inputs* are the distinct requests the guard processed. Development: 1,392 cue targets, plus 464 requests with no supplied cue (no strict target), give 1,856 inputs. Confirmation: 1,872 targets share 1,848 distinct requests. Pass means joint pass; strict-check rows are labelled. Differences use unrounded rates, so a difference of printed levels can differ by 0.1. Right: stored examples, verbatim, by the rule in each row; not typical. Brackets: 95% whole-family bootstrap intervals for paired differences. Repair rates condition on initial strict-check failures; already-correct and pass-to-failure rows on initial strict-check passes.

(family resampling) gives −14.6 [−28.3, −1.6], which excludes zero: exploratory, so the frozen verdict is unchanged, but its margin is thin. Against its own failed original (a paired contrast on 20 pairs), a repair keeps the meaning in 6 of 20

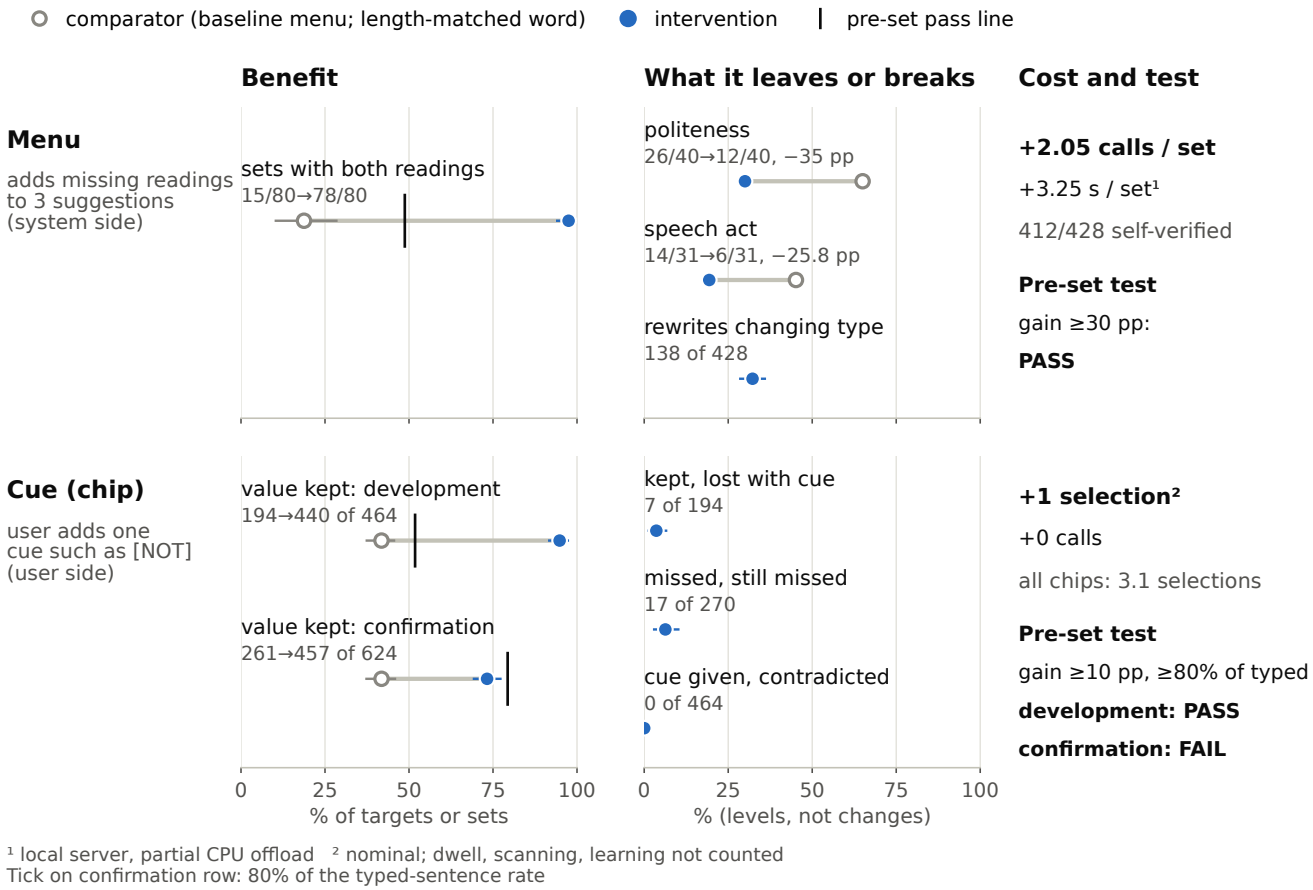


Figure 2: Two add-ons help in one place and cost something elsewhere (machine-extracted labels; four models). Open circle: comparator; filled: intervention; bars: 95% cluster-bootstrap intervals; black tick: the pass line set in advance; pp: percentage points. Plotted values are percent *levels*; changes in points are printed. The cue passes both clauses on the development set but *fails* its ≥80%-of-typed clause on the confirmation set (only its gain clause replicates). The guard is in Table 2.

pairs and the original in none (+30.0 points [9.1, 46.8], exact sign test $p = 0.03$, uncorrected and exploratory; the other 14 pairs are unchanged on meaning), and it removes invented stance (a meaning, certainty, politeness or speaker the input did not give) in 10 of 19 pairs and adds it in none (−52.6 points [−69.1, −26.4], $p = 0.002$). Repairs are not more usable than their originals (paired usable −5.0 points [−20.2, +10.8]).

6.3 Two more ledger rows: the menu and the cue

The complete menu raises coverage of both polarity or certainty readings (+78.75 points, passes its line) but removes the polite option on other dimensions (Figure 1, row 2). The cue keeps the critical value in 94.8% of development targets (97.3% of typed input), and its gain over the length-matched word replicates on 78 fresh speech-act families (31.4 points [25.0, 37.9]), but there it reaches 73.8% of the typed-input rate, below its 80% line (Figure 2). The bare-keyword default is not a prompt artifact. On bare polarity inputs (4 models, both sources), affirmative output among committed answers was 99.5% with the cue legend, 98.4% with the legend removed and 99.9% when the prompt said markers may be missing and not to assume one.

6.4 Realistic-input checks

We ran the pipeline on 547 minimal pairs written by 20 non-author annotators and on keywords that 20 annotators typed for 600 AACText messages. On the human-written pairs the cue gain replicates on all 6 dimensions (35.3 to 82.7 points), and the three confirmation contrasts replicate in direction: speech-act cue +35.3 points, politeness addition +7.6, and a bare affirmative default of 99.8% [99.3, 100.0]. Two controls on these pairs show that the missing cue, not the chip, is the lever. An irrelevant chip from another dimension adds little: the cue chip beats it by 30.0 to 80.9 points by dimension. On these pairs the same cue as a plain word (`not`, `please`) was declared equivalent to the chip (interval within ±5 points) on 3 of 6 dimensions (largest difference 2.1); the chip leads by under 1 point on polarity and time/quantity, and the word beats the chip on speech act by 8.2 points. With 6–7 writers per contrast we claim direction and consistency only: every writer's mean has the same sign, and the replication is

exploratory. In the keyword study about one keyword set in five lost the negation (20.8% [16.9, 25.2]; 21.4% [15.4, 27.8] after resampling messages and reweighting to the sampling frame), on cue-enriched messages that carry a negation.

Defaults: a good guess with an asymmetric cost. When keywords carry no negation word, the models write an affirmative sentence for 99.9% of such human keyword sets. That is the Bayes-preferred guess: 8.6% of AACText messages are negated, and among human keyword sets with no negation word the source was negated only 2.7% [1.8, 3.7] of the time (2.0% from corpus prevalence), against 11.4% for the crude compressor; here p is the probability that the typed source was negated given a keyword set with no negation word, whereas 8.6% is the unconditional share of negated messages, and Section 6.6's 2.6% is the same p on that analysis's 332 exposure units; guessing affirmative is right 97.3% of the time. The harm is the asymmetric cost: about 24 [15, 33] of every 1,000 messages are silently reversed when the user types keywords unaided (a posterior-based estimate from the keyword study; Table 3 lists it with the two other estimates of the same quantity), against 0.8 spurious negations, and the guess breaks even only if a reversal costs under 36 times a spurious negation (the posterior odds $(1-p)/p$). An accuracy metric rewards this behavior, so a ledger puts a default's cost beside its hit rate. Four affirmative rates use different inputs and are not interchangeable: 99.9% of human keyword sets with no negation word; 99.8% of bare human-written pairs; 98.4% to 99.9% of committed answers on bare polarity inputs across the legend arms (four models, one greedy draw each; 99.9% when told not to assume). The default holds on the 6 models from 5 families (at least 99.0% affirmative) and on a lite Gemini model (97.8% and 98.9% on the legend arms). Chen et al. (2023) reported that models write sentences wrongly positive about negative commonsense facts in keyword-to-sentence generation; their measurements are on negative commonsense facts, not AAC messages; we price the default against a human base rate on AAC-style messages and show it survives "do not assume" prompting.

6.5 Two ledger columns from people: writer intent and clinical weight

Writer intent (first test of H1). Each of 20 writers judged 10 outputs generated from their own pairs (200 in all), choosing the version they meant. A cue raises the share matching the intended version from 11.2% to 39.2%, +28.0 points [14.8, 42.0] (writer-clustered; keywords-only is a reference level, not a failure rate). Yet writers accepted 1 of all 200 outputs as written with no edit (a different test from naming the version they meant), and the judge's critical-dimension label agrees with the writer's four-way label at only $\kappa = .24$. Keeping the critical commitment is necessary, not sufficient. Among the 130 of those 200 outputs whose critical value the judge scored correct (126 with a decided answer; 4 unsure), writers rejected 77/126 as not the version meant. A post hoc decomposition (exploratory; classifier agreement with 40 hand-coded items $\kappa = .71$, optimistic because its rules followed those items) finds mainly compression loss and event/role misresolution among the judge-correct outputs writers did not accept: 98% of the writers' sentences contain words no keyword supplied, 49% of the outputs change who does what, and added content is the primary reason in 0 cases but is present in 22% of these outputs (the judge's added-content flag: 22%). Two AAC-user design consultants independently asked that relations (who does what) be preserved.

Estimator	Unit resampled	Silent reversals per 1,000
Posterior (Sec. 6.4)	messages, reweighted to the sampling frame	24 [15, 33]
Model-visible	exposure units	23.9 [16.0, 32.5]
Bare-arm regeneration	same items, regenerated bare (point rate)	23.8

Table 3: Three estimators of one quantity, not independent replications.

Clinical weight. Two speech-language pathologists (expert review of study materials, not a population estimate) rated how serious each of seven kinds of silent meaning change is; both rated a dropped or added negation 5 of 5 and would block it. Weighting each judge-flagged failure by the severity of its type, the guard lowers the clinically weighted failure rate from 22.0% to 14.7% (confirmation) and a supplied cue lowers it on human-written pairs from 57.4% to 14.9% (chip) and 17.8% (word); on selected judge-failed items they judged repairs acceptable to speak in 13/40 rater-judgments against 0/20 for the originals (descriptive, not comparable arms).

6.6 Frontier limits: the cue's measured price, and what asking and uncertainty do not do

These results are provisional: stored outputs, machine labels resolved by a blind arbiter, the three local primary models (Gemma-3-4B, Llama-3.1-8B, Qwen2.5-7B), 332 exposure units. Always committing leaves about 23.9 silent negation reversals per 1,000 messages by the model-visible estimator (Table 3); the abstract and the cue comparison below use the bare-arm regeneration rate, 23.8, because it shares items with the cue arms. Supplying the cue on every message (derived: bare rate times the share not restored, regenerated on the same items) cuts polarity reversals per 1,000 from 23.8 [23.0, 24.3] to 1.6 [0.7, 2.7] (chip) and 2.6 [1.4, 3.9] (word), and politeness reversals from 51.8 to 0.7 and 1.2; the chip and the word are not distinguishable on polarity or speech act on these items; speech-act restoration is 75.6% with a chip and 90.4% with chip plus guard. Ask policies, including a CLAM-style trigger (21.2 reversals per 1,000 against 21.3 for gated random), show no detectable benefit over gated-random asking after Holm correction (power limited, not equivalence), and model sampling uncertainty ranks at-risk inputs at AUROC .56 against .75 for a lexical model. The cue has to come from the user.

H3 study: micro-randomized trial (MRT), pre-specified design; no value is a result	
Estimand	Average causal excursion effect (Boruvka et al. 2018) on a proximal outcome of delivering the regenerated sentence versus the original, at decision points where the monitor flags a conflict, for adult AAC users who compose with LLM keyword expansion
Arms, unit	Randomized per flagged message within user (Klasnja et al. 2015), fixed probabilities: silent regeneration, flag-and-confirm, original; cues can only be offered, so they are a between-user factor
Outcomes	Proximal: sent without edit, one-tap "said what I meant" on a random subset of sends, time to send. Longitudinal: weekly rates, abandonment, reliance, week as moderator
Assumptions	Positivity (every flagged point has a nonzero chance of each arm, by design); consistency (the delivered sentence is the arm); no interference across a user's decision points beyond what we model; carryover is the main threat, so we target the excursion effect (deviate once, then resume usual practice)
Sample size	Illustrative, one two-arm between-user contrast with a cluster design effect, no multiplicity adjustment, typed assumptions to replace with pilot estimates: standardized effect 0.2, within-user correlation 0.3, 40 flagged points per user, 80% power, 5% level, about 125 users. The monitor flagged about 8.5% of constructed inputs (real-use rate unknown), so 40 flagged points need roughly 473 messages per user, a feasibility limit of its own
Analysis	Weighted, centered least squares with user-clustered errors

Table 4: The human-effect study that tests H3.

7 From Ledger to a Pre-Specified Human-Effect Study

Each prediction names its test and what would demote it; H1 and H2 were checked post hoc on data already seen, and the H3 design is pre-specified in version control.

Predictions. H1 (writer intent). Among outputs the scorer counts as preserving a writer's critical value, writers report that the output "said what I meant" more often than among outputs it counts as failing; a gap whose interval includes zero demotes the scored endpoint. This first, post hoc test is a consistency check, close to definitional, and differs from acceptance as written with no edit: writers named the version they meant in 49/126 outputs whose critical value the judge scored correct, against 0/69 of those it scored wrong (Section 6.5). **H2 (repair usability).** A scored repair is usable at no less than 10 points below a never-failed control; the 160-text audit tested this under the frozen rule of Section 6.2, so H2 is unresolved. **H3 (collateral).** Offline changed-correct and coverage loss predict a later drop in users' "said what I meant" rate; if not, evaluation of this kind should be demoted. H3 needs AAC users and is the MRT of Table 4.

Design implications from two AAC-user consultants. Two paid AAC-user design consultants rated our five requirements (keep a button's meaning; do not change typed text; do not quietly guess; a new button must not break what was right; an honest test screen) important, and each asked for a sixth (we report requirement changes only): **R6, a draft is never spoken without the person's approval**, with the original words kept beside the draft and speaking a separate deliberate act. They also asked for an exact-copy default for typed sentences, a visible scope for NOT, and showing where two drafts differ; R6 makes flag-and-confirm the default design.

8 Limitations

Human outcomes and raters. Results are changes in stored text; communication success, agency, effort and trust are untested (Section 7), and the audit's raters are non-author peers, not AAC users. *Labels and scope.* Endpoints use an LLM extractor validated on untreated outputs only, with post-outcome arbitration in the confirmation; strip-only covers one failure type; the menu was not run on the confirmation set; items are constructed, open models are at most 14B and 4-bit, and AACText is crowd-written fiction. *Community involvement.* Two paid AAC-user design consultants and two speech-language pathologists reviewed requirements and study materials (n=2 each; expert review, not a population estimate); the requirements remain our proposals, with no claim of benefit to AAC users. *Ethics.* No data were collected from AAC users. All annotators were adult non-authors recruited from the authors' university peers, identified only by an ID; each consented and was reimbursed for their time. They labelled model outputs and audited items under a fixed codebook, blinded to model, condition and source; some also chose keywords for public AACText messages or wrote new sentence pairs, released under CC BY 4.0 with recorded consent. No personal or health data were collected, and we did not seek an institutional review determination.

9 Conclusion

On the confirmation set's speech-act families the guard raises the joint pass rate by +9.83 points, repairs 52.51% of strict-check failures with 0 of 26 changed texts failing, and beats resampling at matched temperature (+6.94 points). In the human audit, repairs keep the meaning and remove invented stance relative to their failed originals in an exploratory paired contrast on 20 pairs, though they are not more usable than those originals; against never-failed controls the frozen rule returns no detectable cost, the deployment clause is unresolved and stricter endpoints show a cost (Section 6.2). A supplied cue removes nearly all silent reversals on the same items. The ledger records what remains (deletion alone earns +3.47 of the +9.83 points; writers accept 1 of 200 outputs as written), and the pre-specified study turns these into predictions that AAC users can test. Scripts that rebuild every number accompany this submission.

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